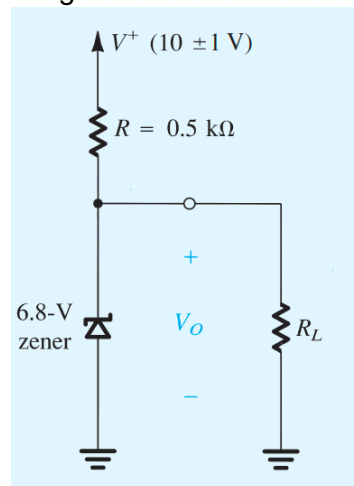
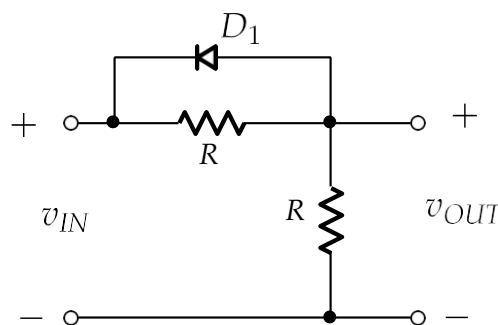


Homework 8 Topic: Diodes II + BJTs

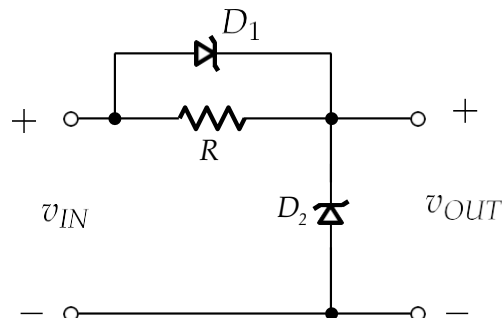
- Consider a half-wave peak rectifier fed with a voltage v_s having a triangular waveform with 24-V peak-to-peak amplitude, zero average, and 1-kHz frequency. Assume that the diode has a 0.7-V drop when conducting. Let the load resistance $R = 100\Omega$ and the filter capacitor $C = 100\mu\text{F}$. Find the average dc output voltage, the time interval during which the diode conducts, the average diode current during conduction, and the maximum diode current.
- Zener Diode Regulator: For this problem you will need the data sheet for a Zener Diode, which you can find your own or use the one Fairchild Semiconductor that is provided on Canvas. Note that the data sheet provides values for V_Z , which is measured at the current I_Z , as well as for the series resistance. We want to design a 12 V regulator using a reverse-bias Zener diode similar to the circuit below the load of $2.2\text{ k}\Omega$. For this design problem, the supply voltage V^+ is $25 \pm 2.5\text{ V}$ (not $10 \pm 1\text{ V}$ as shown in the figure below).
 - Choose one Zener diode to regulate the load voltage to approximately 12 V. Based on the information in the data sheet, find V_{Z0} for this diode.
 - If the Zener current is 25 mA, find V_Z . Then find the current through R , and from that determine a value for R . Use the nominal voltage for the power supply (i.e., 25 V).
 - Find the line regulation.



- For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$ and $r_D = 0$:
 - Write an expression for v_{OUT} in terms of v_{IN} .
 - Sketch a graph of v_{OUT} vs. v_{IN} .



4. Repeat problem 3 with $r_D = R$.
5. For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$, $V_{Z0} = 1.4\text{ V}$, $r_D = r_z = 0$:
 - a. Write an expression for v_{OUT} in terms of v_{IN} .
 - b. Sketch a graph of v_{OUT} vs. v_{IN} .



6. Measurements of V_{BE} and two terminal currents taken on a number of npn transistors operating in the active mode are tabulated below. For each, calculate the missing current value as well as α , β , and I_S as indicated by the table.

Transistor	a	b	c	d	e
V_{BE} (mV)	700	690	580	780	820
I_C (mA)	1.000	1.000		10.10	
I_B (μA)	10		5	120	1050
I_E (mA)		1.020	0.235		75.00
α					
β					
I_S					

7. For the circuits below, assume that the transistors have very large β . Some measurements have been made on these circuits, with the results indicated in the figure. Find the values of the other labeled voltages and currents.

